

Fine-Tuning BiomedParse for Stroke Detection and Segmentation: Comparison with Gemini 2.5 Pro and GPT-5

Artun Gunturkun¹, Halil Ibrahim Gulluk², Ilker Ozgur Koska³

¹ Henry M. Gunn High School, Palo Alto, California, USA

² Electrical Engineering, Stanford University, Palo Alto, California, USA

³ Acibadem Healthcare, İzmir Kent Hospital, Department of Radiology, İzmir, Turkey

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ABSTRACT

Importance: Fast and accurate differentiation between ischemic and hemorrhagic stroke on computed tomography (CT) is crucial for treatment decisions. While deep learning models like CNNs or U-Nets have shown promise, they usually struggle with irregular boundaries of early stroke signs and require large training data. Vision language models, which combine semantic medical knowledge with visual analysis, may overcome these challenges.

Objective: To fine-tune and evaluate BiomedParse [1], a visual language model, for the automated detection of ischemic and hemorrhagic stroke lesions and compare it with Gemini 2.5 Pro and GPT 5.

Design, Setting, and Participants: This retrospective study utilized the Teknofest-2021 Stroke Data Set [2], consisting of 6,650 CT slices (1,130 ischemic, 1,093 hemorrhagic,

and 4,427 healthy) collected from the Turkish Ministry of Health's e-Pulse system (2019-2020). The dataset was split into training and testing sets. We fine-tuned the BiomedParse model using 2D CT slices, their corresponding masks and prompts (e.g., "stroke is present", "bleeding is present", "normal and healthy"). We then asked the fine-tuned model two questions: "Is there a stroke in the image?" and "Is there bleeding in the image?" We asked the same questions to Gemini 2.5 Pro and GPT-5.

Main Outcomes and Measures: Primary outcomes were stroke and bleeding detection, assessed using classification metrics such as accuracy, sensitivity, specificity and F1 score.

Results: The fine-tuned BiomedParse model demonstrated higher performance compared with Gemini 2.5 Pro and GPT-5 across all classification metrics. For stroke detection, on the test set, fine-tuned BiomedParse (**Figure 1**) achieved higher accuracy (95.2% vs 62.8% and 78.4%), F1 score (85.8% vs 35.6% and 4.0%), sensitivity (85.4% vs 60.6% and 2.7%), and specificity (97.2% vs 63.3% and 93.9%) compared with Gemini 2.5 Pro and GPT-5, respectively. In bleeding detection, performance differences were more significant: fine-tuned BiomedParse demonstrated almost perfect classification, with an accuracy of 98.9%, F1 score of 96.5%, sensitivity of 95.4%, and specificity of 99.6% , whereas both Gemini 2.5 Pro and GPT-5 showed lower sensitivity (2.3%, 9.1%), F1 scores (2.7%, 15.8%) and high specificity (86.7%, 98.7%), respectively. Higher specificity and lower sensitivity suggests that LLMs have a

tendency to label these images as healthy either because they couldn't detect stroke/bleeding or they don't truly understand the underlying findings in the images.

Conclusions: Fine-tuning a biomedical foundation model provides a scalable and high performing alternative to manual annotation. LLMs still require domain specific adjustment to perform adequately on medical images.

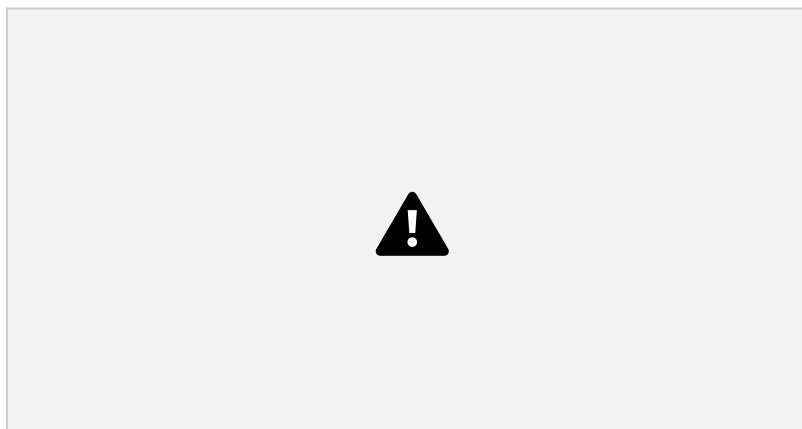
References:

[1] Zhao, T., Gu, Y., Yang, J. *et al.* A foundation model for joint segmentation, detection and recognition of biomedical objects across nine modalities. *Nat Methods* 22, 166–176 (2025).

<https://doi.org/10.1038/s41592-024-02499-w>

[2] Koç, U., Akçapınar, S. E., Özkaya, Y. A., *et al.* Artificial Intelligence in Healthcare Competition (TEKNOFEST-2021): Stroke Data Set. *Eurasian J Med.* 54(3):248-258 (2022) doi: 10.5152/eurasianjmed.2022.22096. PMID: 35943079; PMCID: PMC9797774.

Figure 1: Comparison of segmented stroke regions of fine-tuned model versus ground truth



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This retrospective study aimed to fine-tune and evaluate BiomedParse, a vision-language model, for automated detection of ischemic and hemorrhagic stroke lesions, and to compare its performance with Gemini 2.5 Pro and GPT-5. We used the Teknofest-2021 Stroke Dataset, which includes 6,650 CT slices (1,130 ischemic, 1,093 hemorrhagic, and 4,427 healthy) collected from the Turkish Ministry of Health's e-Pulse system (2019–2020). The dataset was split into training and testing sets. BiomedParse was fine-tuned using 2D CT images, corresponding segmentation masks, and prompts such as “stroke is present,” “bleeding is present,” and “normal and healthy.” Each model was then asked two questions: “Is there a stroke in the image?” and “Is there bleeding in the image?”

Performance was evaluated using accuracy, sensitivity, specificity, and F1 score. For stroke detection, BiomedParse achieved higher accuracy (95.2% vs 62.8% and 78.4%),

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